

SOFTWARE ENGINEERING LAB 01

REQUIREMENTS TABLE :

Problem Statement #56 :- Peer Skill Exchange & Mentorship Network (PES1UG24CS244)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall maintain a time-credit balance for each member, crediting 1 hour upon verified completion of a teaching session.	High	Pass: Time credit transfers from learner to mentor after session sign-off. Fail: Member with 0 credits books a session.	Core mechanism that makes fair, money-free skill trading possible.	Given in problem statement
FR-002	Functional	The system shall allow a Learner Member to search and filter Skill Mentors by skill category, availability, and rating.	High	Pass: Searching "Guitar" returns only mentors who teach guitar and are available in the next 7 days. Fail: Search returns unrelated or unavailable mentors.	Members must be able to find a suitable mentor quickly, or the exchange never starts.	
FR-003	Functional	The system shall allow a Learner Member to book a session with a Skill Mentor only if the learner holds at least 1 hour of time-credit.	High	Pass: Booking succeeds and the slot is reserved when balance >= 1 hour. Fail: Booking is rejected and an "insufficient credit" message is shown when balance < 1 hour.	Prevents members from taking mentoring time they cannot pay back in kind.	
FR-004	Functional	The system shall require both the Learner Member and the Skill Mentor to confirm (sign-off) that a session took place before any time-credit is transferred.	Medium	Pass: Credit moves only after both parties tap "Confirm Completion." Fail: Credit transfers when only one party has confirmed.	Mutual sign-off stops one party from falsely claiming a session happened.	
FR-005	Functional	The system shall allow members to submit a 1-5 star rating and a written comment for their exchange partner within 7 days of a completed session.	Medium	Pass: A rating submitted within 7 days is saved and shown on the partner's profile. Fail: A rating attempt after 7 days is blocked with an "expired" message.	Feedback ratings build trust and let good mentors stand out in search results.	
NFR-001	Nonfunctional	The session scheduling calendar must sync seamlessly with Google Calendar and Outlook via iCal export feeds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Members already live in their personal calendars; the platform must fit into that habit, not fight it.	Given in problem statement
NFR-002	Nonfunctional	The system shall encrypt all stored personal data and in-transit communication (profile details, messages, ratings) using TLS 1.2+ and AES-256 at rest.	High	Pass: A security scan/penetration test finds no personal data stored or transmitted in plain text. Fail: Any personal data is found unencrypted in storage or transit.	The platform handles personal contact and scheduling data, so it must protect member privacy and meet basic security expectations.	